Chemistry – 100% AI SLOPPED!
HSLU, Semester 5

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Contents

Scope and notation **3**

1 Substances and mixtures **3**
1.1 Classification of matter . . . 3
1.2 Physical, chemical and nuclear changes . . . 4
1.3 Separation methods . . . 4
1.4 States of matter and phase transitions . . . 5
1.5 Reading a phase diagram . . . 5
1.6 Density and the anomaly of water . . . 6

2 Amount of substance and solutions **7**
2.1 Atomic, molecular and molar mass . . . 7
2.2 The mole and Avogadro constant . . . 7
2.3 Solutions and solubility . . . 7
2.4 Concentration units . . . 8
2.5 Preparing and diluting solutions . . . 8

3 Atomic structure and radioactivity **9**
3.1 The nucleus and electron cloud . . . 9
3.2 Nuclide notation and ions . . . 9
3.3 Isotopes and isobars . . . 9
3.4 Radioactive decay . . . 9
3.5 Half-life and activity . . . 10
3.6 Radiocarbon dating . . . 10
3.7 Fission, fusion and energy . . . 10

4 Atomic models and electron configuration **11**
4.1 Development of the atomic model . . . 11
4.2 Orbitals and quantum numbers . . . 11
4.3 Filling rules . . . 12
4.4 Writing a configuration . . . 12

5 The periodic table **13**
5.1 Periods, groups and blocks . . . 13
5.2 Metals, nonmetals and metalloids . . . 13
5.3 Periodic trends . . . 13
5.4 Oxidation numbers . . . 14
5.5 Abundance of elements . . . 14

6 Chemical bonds and hybridization **15**
6.1 Why atoms bond . . . 15
6.2 Octet rule and its limits . . . 15
6.3 Covalent, ionic and metallic bonding . . . 15
6.4 Bond polarity and molecular polarity . . . 16

6.5 Sigma and pi bonds . . . 16
6.6 Hybridization . . . 16
6.7 Coordination compounds . . . 17
6.8 Intermolecular interactions and properties . . . 17

7 Lewis structures and molecular geometry 19
7.1 Lewis symbols . . . 19
7.2 Procedure for drawing a Lewis structure . . . 19
7.3 Formal charge . . . 19
7.4 Resonance and delocalization . . . 20
7.5 The VSEPR model . . . 20
7.6 Geometry reference . . . 20
7.7 Molecular drawings . . . 21

8 Chemical reactions and stoichiometry 22
8.1 Representing a reaction . . . 22
8.2 Balancing equations . . . 22
8.3 Using mole ratios . . . 22
8.4 Recognizing reactions . . . 22

9 Thermodynamics and ideal gases 23
9.1 System and surroundings . . . 23
9.2 First law and sign convention . . . 23
9.3 Ideal-gas equation and gas laws . . . 23
9.4 Molar gas volume . . . 24
9.5 Enthalpy and reaction heat . . . 24
9.6 Entropy and the second law . . . 24
9.7 Gibbs energy and spontaneity . . . 24
9.8 Calculations from standard data . . . 25

10 Chemical equilibrium and kinetics 26
10.1 Dynamic equilibrium . . . 26
10.2 Reaction quotient and direction . . . 26
10.3 Le Chatelier’s principle . . . 26
10.4 Reaction rate . . . 27
10.5 Activation energy and catalysts . . . 27

A Acid–base reference from the supplied sheets 28
A.1 Acids, bases and conjugate pairs . . . 28
A.2 Water, pH and pOH . . . 28
A.3 Acid and base constants . . . 28
A.4 Useful pH approximations . . . 28
A.5 Buffers . . . 29
A.6 Selected acid data . . . 29

B Redox and electrochemical reference 30
B.1 Oxidation and reduction . . . 30
B.2 Balancing redox equations by half-reactions . . . 30
B.3 Standard reduction potentials . . . 30
B.4 Cells and the Nernst equation . . . 31

C Formula overview 32

D Source map and conventions 33

Scope and notation

These notes follow the theory material in `TA.BA_CHEM_MM.H2602`: introduction to matter, atomic structure and radioactivity, atomic models, the periodic table, electron configurations, chemical bonds, Lewis structures, and chemical reactions (weeks 1–6). Exercise sheets, solution sheets and exercise prompts inside the lectures are excluded. Small molecular illustrations are included to explain the theory.

The supplied formula sheet and MEP reference notes also contain acid–base and electrochemical material. This is collected in Appendices A and B; it is reference material, not a reconstruction of lecture weeks that are absent from the folder. A source map is provided in Appendix D.

Symbol	Meaning	Symbol	Meaning
m	Mass	n	Amount of substance
N	Number of specified particles	N_A	Avogadro constant
M	Molar mass	c	Amount concentration (molarity)
V	Volume	ρ	Density
Z	Atomic number	A	Mass number
T	Absolute temperature in kelvin	p	Absolute pressure

The lecture material sometimes uses M for molarity. Here M always denotes molar mass and c denotes molarity, to keep the two quantities distinct. Chemical formulas and equations use `mhchem`; molecular drawings use `chemfig`. A line represents a shared electron pair and a pair of dots represents a lone pair.

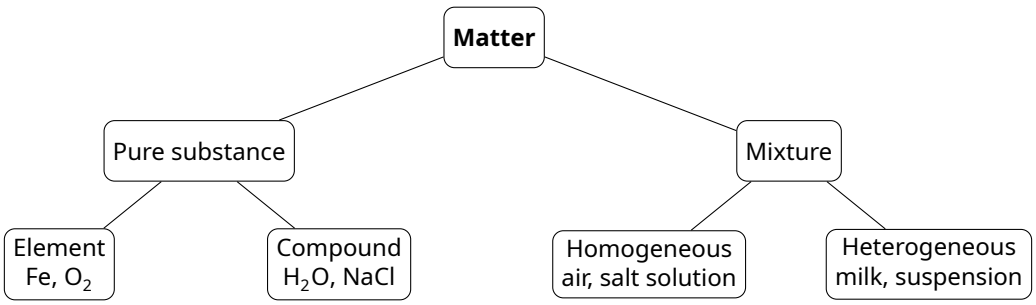
1 Substances and mixtures

Course source: SW1, Introduction to Substances.

1.1 Classification of matter

Matter has mass and occupies space. Its classification depends on composition and on whether its properties are uniform throughout the sample.

- A **pure substance** has a fixed chemical composition and characteristic properties at specified temperature and pressure.
- An **element** contains atoms with the same atomic number and cannot be decomposed into simpler substances by chemical methods.
- A **compound** contains two or more elements in a fixed ratio. Chemical reactions can decompose it into other substances.
- A **mixture** contains several substances in variable proportions. Its components can be separated using differences in physical properties.



A **homogeneous mixture** is uniform on the scale considered and forms one phase. A **heterogeneous mixture** contains distinguishable phases; these need not be visible to the naked eye. Milk, for example, is an emulsion. A **phase** is a region with uniform physical and chemical properties.

Mixture	Dispersed substance and medium	Example
Solution	Molecular or ionic mixing in one phase	Salt water; air
Suspension	Solid particles in a liquid	Muddy water
Emulsion	Liquid droplets in another liquid	Milk; oil droplets in water
Aerosol	Liquid droplets or solid particles in a gas	Fog; smoke
Solid mixture	Several solid phases	Granite

1.2 Physical, chemical and nuclear changes

1. A **physical change** alters the state, shape, size or distribution of substances without changing their chemical identity. Melting ice and separating a mixture are physical processes.
2. A **chemical reaction** forms different substances through a rearrangement of atoms and electrons. Chemical bonds are broken and formed; atomic nuclei retain their identities.
3. A **nuclear reaction** changes a nucleus. It may change the element and involves much larger characteristic energies per event than chemical reactions.

Distinguishing criterion: ask whether new substances form. Reversibility alone does not distinguish physical from chemical changes. Breaking glass is physical, while a chemical reaction can be reversible.

Physical properties include density, melting temperature and conductivity. Chemical properties describe reactivity, such as the tendency to burn or oxidize. An **extensive** quantity depends on sample size (mass, volume); an **intensive** quantity does not (temperature, density of a uniform phase).

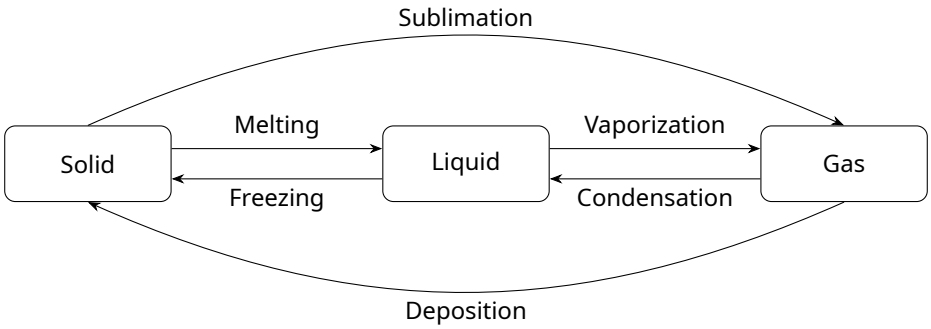
1.3 Separation methods

Method	Property used	Principle or application
Sedimentation	Density and particle settling	Suspended solids settle under gravity; used in wastewater treatment.
Decantation	Separated phases	Pour off a liquid above a settled solid, or separate immiscible liquid layers using a separating funnel.
Filtration	Particle size	A porous medium retains the residue; the fluid passing through is the filtrate. Dissolved ions pass through ordinary filters.
Distillation	Relative volatility	Vaporize and condense a mixture to obtain a distillate enriched in more volatile components.
Extraction	Different solubilities	Transfer a substance into a suitable solvent; for example, extracting soluble compounds from tea.
Adsorption	Affinity for a surface	Substances bind to a solid surface, as in activated-carbon treatment.
Absorption	Uptake into a bulk phase	A gas dissolves in a liquid absorbent, as in gas scrubbing.

Adsorption occurs at a surface; **absorption** occurs throughout the absorbing phase. In practice, several separation steps are often combined.

1.4 States of matter and phase transitions

State	Shape	Volume	Particle picture
Solid	Approximately fixed	Approximately fixed	Particles vibrate near fixed positions.
Liquid	Container shape	Approximately fixed	Closely packed particles can move past one another.
Gas	Fills container	Container volume	Widely separated particles move freely.
Plasma	No fixed shape	No fixed volume	Contains mobile ions and electrons; conducts electricity.



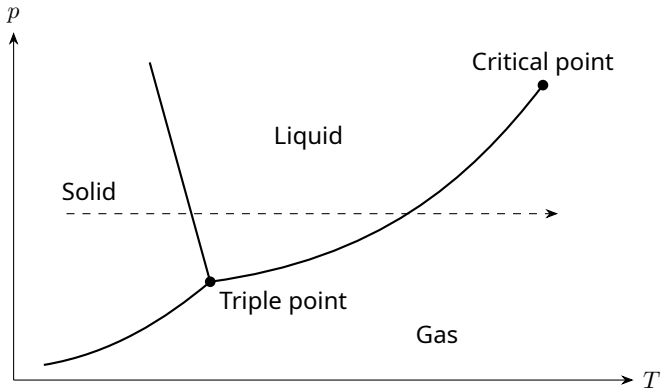
Melting, vaporization and sublimation absorb heat; the reverse processes release heat. During a phase transition of a pure substance at constant pressure, heat changes the phase fraction while the temperature remains constant.

1.5 Reading a phase diagram

A phase diagram shows the stable phase as a function of pressure and temperature. Inside a region, one phase is stable; on a boundary, two phases coexist.

- At the **triple point**, solid, liquid and gas coexist.
- At the **critical point**, the liquid–gas coexistence curve ends. Above the critical temperature, pressure alone cannot produce a distinct liquid phase.
- Heating at constant pressure follows a horizontal path on a p–T diagram. Compression at constant temperature follows a vertical path.

For water near atmospheric pressure, increasing pressure lowers the melting temperature because liquid water is denser than ordinary ice. This gives the solid–liquid boundary a negative slope in that region.



Schematic phase diagram for water near the familiar phase boundaries; not to scale. The dashed path illustrates heating at fixed pressure.

$$\boxed{\rho = \frac{m}{V}, \qquad m = \rho V}$$

1.6 Density and the anomaly of water

Common units are kg m^{-3} and g cm^{-3}: $1\,\mathrm{g\,cm^{-3}} = 1000\,\mathrm{kg\,m^{-3}}$. Density depends on temperature and pressure, especially for gases. At ordinary pressure, liquid water has its maximum density near $4\,^{\circ}\mathrm{C}$. Ordinary ice is less dense than liquid water and therefore floats. Its open hydrogen-bonded structure explains the expansion on freezing.

2 Amount of substance and solutions

Course source: SW1 and the supplied formula sheet, items 1–4.

2.1 Atomic, molecular and molar mass

The **atomic mass constant** m_{u} is one twelfth of the mass of a free carbon-12 atom in its ground state and at rest. The unified atomic mass unit is $1\,\mathsf{u} = m_{\mathsf{u}} \approx 1.66054 \times 10^{-27}\,\mathsf{kg}$.

The **relative atomic mass** A_{r} is dimensionless:

$$A_{\mathsf{r}} = \frac{\overline{m}_{\mathsf{atom}}}{m_{\mathsf{u}}}, \qquad \overline{m}_{\mathsf{atom}} = \sum_i x_i m_i, \qquad \sum_i x_i = 1.$$

Here x_i is the abundance fraction of isotope i, and m_i its atomic mass. The periodic table usually gives an abundance-weighted atomic weight, not the integer mass number of a particular isotope.

For a molecule, add the atomic masses with their formula subscripts as multipliers. For an ionic solid, the corresponding quantity is a **formula mass**, since the solid is a lattice rather than a collection of isolated molecules. For example, $M(\mathsf{H_2O}) = 2M(\mathsf{H}) + M(\mathsf{O}) \approx 18.015\,\mathsf{g\,mol^{-1}}$. At the precision used in this course,

$$\boxed{M(\mathsf{A}_p\mathsf{B}_q\mathsf{C}_r) = pM(\mathsf{A}) + qM(\mathsf{B}) + rM(\mathsf{C})}$$

the numerical mass of a particle in u equals the numerical molar mass in $\mathsf{g\,mol^{-1}}$.

2.2 The mole and Avogadro constant

The **mole** measures the amount of specified entities: atoms, molecules, ions, electrons or formula units. One mole contains exactly $6.02214076 \times 10^{23}$ such entities (the SI definition; see Appendix D). Always

$$\boxed{N_{\mathsf{A}} = 6.02214076 \times 10^{23}\,\mathsf{mol^{-1}}, \qquad n = \frac{N}{N_{\mathsf{A}}} = \frac{m}{M}, \qquad N = nN_{\mathsf{A}}}$$

specify what is being counted: one mole of $\mathsf{O_2}$ molecules contains two moles of oxygen atoms. For NaCl, count formula units or the individual ions, according to the question.

2.3 Solutions and solubility

A **solution** is a homogeneous mixture. The **solvent** is the dissolving medium, commonly the component present in the largest amount. A **solute** is a dissolved component. Solvent molecules surround dissolved particles; this is **solvation**, or **hydration** when the solvent is water.

Solubility is the equilibrium amount of a solute that can dissolve under specified conditions. State the temperature, pressure where relevant, and the concentration basis (for example, grams per litre of solution or per 100 g of solvent).

- **Unsaturated:** more solute can dissolve.
- **Saturated:** the dissolved concentration equals the solubility; with excess solid present, dissolution and crystallization balance.
- **Supersaturated:** the concentration exceeds the equilibrium value. Such a solution is metastable and can crystallize after a disturbance or seeding.

2.3.1 Energy of dissolution

Dissolution requires separating solute particles and rearranging solvent interactions, while forming solute–solvent interactions releases energy.

$$\Delta H_{\mathsf{sol}} = \Delta H_{\mathsf{separate\ solute}} + \Delta H_{\mathsf{rearrange\ solvent}} + \Delta H_{\mathsf{solvation}}.$$

The net process can be exothermic or endothermic. Solubility is governed by the free-energy balance, not only by the speed of dissolution. Heating often increases the solubility of solids, but this is not a universal rule.

2.3.2 Gas solubility

For a dilute solution of a gas that does not react significantly with the solvent, Henry's law can be written Here p_{gas} is the gas's **partial pressure** above the solution; the units and value of k_{H} depend on the chosen

$$\boxed{c_{\text{gas}} = k_{\text{H}} p_{\text{gas}} \quad \text{(fixed temperature)}}$$

convention. Gas solubility generally increases with partial pressure and decreases with temperature. A pressure decrease can therefore release dissolved gas as bubbles.

2.4 Concentration units

Quantity	Definition	Typical unit or convention
Molarity	$c = n_{\text{solute}}/V_{\text{solution}}$	mol L^{-1}
Mass concentration	$\beta = m_{\text{solute}}/V_{\text{solution}}$	g L^{-1}
Mass fraction	$w = m_{\text{solute}}/m_{\text{solution}}$	Dimensionless; mass percentage $= 100w\,\%$
Volume percentage	$100V_{\text{solute}}/V_{\text{solution}}$	$\%$ by volume; specify temperature
Mass/volume percentage	Grams of solute per 100 mL of solution	$\%\,(m/V)$; not dimensionless
Parts per million	$10^6 w$ on a mass basis	$1\,\text{ppm} = 1\,\text{mg kg}^{-1}$
Parts per billion	$10^9 w$ on a mass basis	$1\,\text{ppb} = 1\,\mu\text{g kg}^{-1}$

For dilute aqueous solutions with $\rho \approx 1\,\text{kg L}^{-1}$, $1\,\text{mg L}^{-1} \approx 1\,\text{ppm}$ on a mass basis. This equivalence is an approximation and does not apply to arbitrary liquids or gas mixtures. Use consistent units: if ρ is in

$$\boxed{c = \frac{m}{MV}, \qquad \beta = cM, \qquad c = \frac{w\rho}{M}}$$

g L^{-1} and M in g mol^{-1}, the last expression gives c in mol L^{-1}. The denominator in molarity is the volume of the **final solution**, not the volume of solvent initially added.

2.5 Preparing and diluting solutions

To prepare a solution from a solid, calculate $m = cMV$, dissolve the weighed solute in part of the solvent, then make up to the required final volume. For a dilution, the solute amount is conserved if no reaction or loss occurs: V_1 is the stock-solution aliquot and V_2 the final diluted volume. Adding solvent lowers

$$\boxed{n_1 = n_2 \quad \Longrightarrow \quad c_1 V_1 = c_2 V_2, \qquad V_1 = \frac{c_2 V_2}{c_1}}$$

concentration, but does not change the amount of solute. Volumes are not always exactly additive when different liquids are mixed.

Keep separate: concentration describes how much solute is present; solubility describes the equilibrium limit. A concentrated solution need not be saturated, and a saturated solution can have a very low concentration.

3 Atomic structure and radioactivity

Course source: SW2, Structure of atoms.

3.1 The nucleus and electron cloud

An atom contains a small, massive nucleus and an electron cloud that determines most of its size. Typical atomic dimensions are of order 10^{-10} m, whereas nuclear dimensions are of order 10^{-15}–10^{-14} m.

Particle	Charge	Approximate mass	Location
Proton	$+e$	$1.0073\,\mathrm{u}$	Nucleus
Neutron	0	$1.0087\,\mathrm{u}$	Nucleus
Electron	$-e$	$0.0005486\,\mathrm{u}$	Electron cloud

Almost all atomic mass is in the nucleus. The number of protons identifies the element; electron arrangement is central to its chemical behaviour.

3.2 Nuclide notation and ions

$$\boxed{{}^{A}_{Z}\mathrm{X}^{q}, \qquad A = Z + N_{\mathrm{n}}, \qquad N_{\mathrm{n}} = A - Z, \qquad N_{\mathrm{e}} = Z - q}$$

Here q is the signed charge number ($+2$ for a doubly positive ion), N_{n} the neutron count and N_{e} the electron count. For a neutral atom, $N_{\mathrm{e}} = Z$.

- A **cation** forms by electron loss and has positive charge.
- An **anion** forms by electron gain and has negative charge.
- Forming an ion does not change the number of protons or the element.

$$\mathrm{Na} \longrightarrow \mathrm{Na}^{+} + \mathrm{e}^{-}, \qquad \mathrm{Cl} + \mathrm{e}^{-} \longrightarrow \mathrm{Cl}^{-}.$$

3.3 Isotopes and isobars

Isotopes have the same Z but different neutron counts and mass numbers. For example, ${}^{12}_{6}\mathrm{C}$, ${}^{13}_{6}\mathrm{C}$ and ${}^{14}_{6}\mathrm{C}$ contain 6, 7 and 8 neutrons. **Isobars** have the same mass number A but different atomic numbers.

Hydrogen isotopes have special names: **protium** ${}^{1}_{1}\mathrm{H}$, **deuterium** ${}^{2}_{1}\mathrm{H}$ and **tritium** ${}^{3}_{1}\mathrm{H}$. Isotopes have similar electronic structures and broadly similar chemistry, but their different masses can affect reaction rates and physical properties. Some isotopes are radioactive, while others are stable.

3.4 Radioactive decay

An unstable nucleus can transform spontaneously and emit ionizing radiation. Nuclear stability depends on both size and neutron-to-proton ratio; light stable nuclei often have comparable neutron and proton counts, while heavier stable nuclei require a neutron excess.

3.4.1 Alpha decay

The emitted alpha particle is a helium nucleus containing two protons and two neutrons: A decreases

$$\boxed{{}^{A}_{Z}\mathrm{X} \longrightarrow {}^{A-4}_{Z-2}\mathrm{Y} + {}^{4}_{2}\mathrm{He}}$$

by four and Z by two. Alpha radiation has high ionization density and a short range in matter.

3.4.2 Beta decay

In **beta-minus decay**, a neutron becomes a proton, emitting an electron and an electron antineutrino:

$$\mathrm{n} \longrightarrow \mathrm{p} + \mathrm{e}^{-} + \overline{\nu}_{\mathrm{e}}, \qquad {}^{A}_{Z}\mathrm{X} \longrightarrow {}^{A}_{Z+1}\mathrm{Y} + {}^{0}_{-1}\mathrm{e} + \overline{\nu}_{\mathrm{e}}.$$

A is unchanged and Z increases by one. The emitted electron is produced in the decay; it is not an orbital electron leaving the atom. In **beta-plus decay**, a proton becomes a neutron with emission of a positron and a neutrino, so Z decreases by one while A stays unchanged.

3.4.3 Gamma emission

An excited nucleus loses energy by emitting a photon: Neither A nor Z changes. Gamma emission often

$$\boxed{{}^{A}_{Z}\mathsf{X}^{*} \longrightarrow {}^{A}_{Z}\mathsf{X} + \gamma}$$

follows an alpha or beta decay.

Radiation	Nature	Qualitative penetration
α	Helium nuclei	Short range; stopped by paper or the outer skin layer.
β	Electrons or positrons	More penetrating than alpha; range depends strongly on energy.
γ	High-energy photons	Highly penetrating; attenuated by substantial thicknesses of dense material.

These are qualitative comparisons, not shielding specifications. The effect of radiation depends on its energy, exposure and whether radioactive material is outside or inside an object or organism. Radon is an example of a radioactive gas that can migrate from the ground into buildings.

3.5 Half-life and activity

For a single radionuclide with no replenishment, the decay rate is proportional to the number of undecayed nuclei: The **activity** $\mathcal{A}$ is the number of decays per unit time:

$$\boxed{N(t) = N_0 e^{-\lambda t} = N_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}, \qquad t_{1/2} = \frac{\ln 2}{\lambda}}$$

$$\mathcal{A}(t) = -\frac{\mathrm{d}N}{\mathrm{d}t} = \lambda N(t), \qquad 1\,\mathsf{Bq} = 1\,\mathsf{s}^{-1}.$$

After one, two and three half-lives, $1/2$, $1/4$ and $1/8$ of the initial nuclei remain. For the same number of radioactive nuclei, a shorter half-life means a higher activity. For the same isotope, activity and remaining radioactive mass decrease by the same factor.

3.6 Radiocarbon dating

Living organisms exchange carbon with their environment. After death, this exchange ceases and the carbon-14 content decreases through beta-minus decay. Using its approximate half-life of 5730 years, The simple relation assumes a known initial carbon-14 ratio and a closed sample. Practical dating also

$$\boxed{t = \frac{t_{1/2}}{\ln 2} \ln\!\left(\frac{N_0}{N(t)}\right) = \frac{t_{1/2}}{\ln 2} \ln\!\left(\frac{\mathcal{A}_0}{\mathcal{A}(t)}\right)}$$

requires calibration and correction for contamination.

3.7 Fission, fusion and energy

Fission splits a heavy nucleus into lighter nuclei, commonly releasing neutrons that can trigger further fissions. A controlled chain reaction produces heat in a nuclear reactor. **Fusion** combines light nuclei and powers stars. Energy release is associated with an increase in nuclear binding energy: Here c_0 is

$$\boxed{E_{\mathsf{released}} = (m_{\mathsf{initial}} - m_{\mathsf{final}})c_0^2}$$

the speed of light, distinct from concentration c. Balance mass numbers and atomic numbers in nuclear equations. Chemical reactions rearrange electrons; nuclear reactions alter the nuclei themselves. Applications discussed in the course include energy production, radiographic inspection, tracers and radiation-assisted pollutant treatment.

4 Atomic models and electron configuration

Course source: SW3, Atomic Models (self-study) and Electronic Configuration.

4.1 Development of the atomic model

Model	Main idea or observation	Limitation or development
Dalton	Matter consists of atoms; chemical reactions rearrange them in simple ratios.	Atoms are not indivisible, and isotopes of an element have different masses.
Thomson	Cathode rays reveal electrons; negative particles are embedded in a positive background.	Does not account for the concentrated atomic nucleus.
Rutherford	Most alpha particles pass through gold foil, while a few deflect strongly: a tiny positive nucleus contains most mass.	Classical orbiting electrons do not explain stable atoms and discrete spectra.
Bohr	Electrons occupy discrete energy levels; transitions absorb or emit light.	Useful for hydrogen-like atoms, but insufficient for general many-electron atoms.
Quantum model	Electrons are described by wavefunctions and orbitals rather than fixed trajectories.	Probability distributions replace classical paths.

For an electronic transition, the photon energy is

$$E_{\text{photon}} = h\nu = \frac{hc_0}{\lambda_{\text{light}}} = |E_{\text{final}} - E_{\text{initial}}|.$$

An upward transition absorbs energy; a downward transition emits it. Characteristic energy differences explain line spectra and flame colours.

4.2 Orbitals and quantum numbers

An **orbital** is a one-electron wavefunction. Its squared magnitude $|\psi|^2$ gives a probability density; an orbital drawing usually encloses a chosen fraction of that probability. An orbital is not an electron's path. The uncertainty principle constrains position and momentum simultaneously:

$$\Delta x \, \Delta p_x \geq \frac{\hbar}{2}.$$

Quantum number	Allowed values	Meaning
Principal, n	$1, 2, 3, \ldots$	Shell; characteristic size and energy scale.
Angular momentum, ℓ	$0, 1, \ldots, n-1$	Subshell and orbital shape.
Magnetic, m_ℓ	$-\ell, \ldots, 0, \ldots, +\ell$	Orbital orientation; $2\ell + 1$ values.
Spin projection, m_s	$+\frac{1}{2}$ or $-\frac{1}{2}$	Two possible spin projections for an electron.

Spin is an intrinsic quantum property, not a literal classical rotation of the electron. The electron has spin quantum number $s = 1/2$; the two alternatives above are the projections m_s.

Subshell	ℓ	Orbitals	Shape (schematic description)	Maximum electrons
s	0	1	Spherical	2
p	1	3	Two lobes	6
d	2	5	Mostly four lobes	10
f	3	7	More complex angular shapes	14

Orbitals in shell $n = n^2$,	maximum electrons in shell $n = 2n^2$

The first four shells have capacities 2, 8, 18 and 32. These are capacities, not the order in which the ground-state atom fills the subshells.

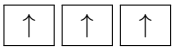
4.3 Filling rules

1. **Aufbau principle:** occupy lower-energy orbitals first. The usual order is summarized below; some atoms have exceptions.
2. **Pauli exclusion principle:** no two electrons in an atom have the same four quantum numbers. Each orbital holds at most two electrons, with opposite spins.
3. **Hund's rule:** occupy degenerate orbitals singly with parallel spins before pairing electrons.

$$\begin{array}{c} 1\mathsf{s} \to 2\mathsf{s} \to 2\mathsf{p} \to 3\mathsf{s} \to 3\mathsf{p} \to 4\mathsf{s} \to 3\mathsf{d} \to 4\mathsf{p} \to 5\mathsf{s} \\ \to 4\mathsf{d} \to 5\mathsf{p} \to 6\mathsf{s} \to 4\mathsf{f} \to 5\mathsf{d} \to 6\mathsf{p} \to 7\mathsf{s} \to 5\mathsf{f} \to 6\mathsf{d} \to 7\mathsf{p} \end{array}$$

The $(n + \ell)$ rule is a useful guide: fill smaller $n + \ell$ first, with smaller n breaking ties. Actual subshell energies also depend on occupancy.

A p^3 subshell illustrates Hund's rule:

 rather than 

For p^4, one pair is necessary:

 .

4.4 Writing a configuration

Determine the number of electrons, fill subshells in order, and check that the superscripts sum to the electron count. A noble-gas symbol in brackets replaces the filled core. For example,

$$\mathsf{Si}: \quad 1\mathsf{s}^2\,2\mathsf{s}^2\,2\mathsf{p}^6\,3\mathsf{s}^2\,3\mathsf{p}^2 = [\mathsf{Ne}]\,3\mathsf{s}^2\,3\mathsf{p}^2.$$

For main-group atoms, the outermost s and p electrons are the **valence electrons**. Core electrons occupy the inner filled shells. Transition-metal bonding can also involve the $(n-1)$d electrons.

When forming a cation, remove electrons from the outermost occupied shell first. For transition-metal cations, the ns electrons are normally removed before $(n-1)$d electrons, even though ns was filled first. Anions gain electrons subject to the same occupancy rules. Configurations with nearby s and d energies can depart from the simple filling sequence; use the supplied periodic-table data when such exceptions are relevant.

5 The periodic table

Course source: SW3, Periodic Table; MEP notes, oxidation-number reference.

5.1 Periods, groups and blocks

Elements are ordered by increasing **atomic number** Z. **Periods** are the seven horizontal rows; **groups** are the 18 vertical columns. Main-group elements in the same group have similar valence-shell configurations, which helps explain their related chemical properties.

The s, p, d and f blocks identify the subshell being populated across a series. For main-group atoms, the period corresponds to the highest occupied principal shell in the usual ground-state configuration. The group number gives the valence-electron count for groups 1 and 2, and group number minus ten for groups 13–18. Helium is the exception: its full shell is $1s^2$.

Family	Outer configuration	Characteristic behaviour
Group 1: alkali metals	ns^1	Easily lose one electron; usually form $+1$ ions. Reactivity generally increases down the group. Hydrogen is not an alkali metal.
Group 2: alkaline-earth metals	ns^2	Usually form $+2$ ions; reducing metals, generally less reactive than neighbouring alkali metals.
Group 17: halogens	ns^2np^5	Commonly form -1 ions; F_2, Cl_2, Br_2, I_2 are diatomic. Oxidizing ability generally decreases down the group.
Group 18: noble gases	ns^2np^6	Filled valence shell (He: $1s^2$); very low reactivity, but not universally incapable of bonding.

5.2 Metals, nonmetals and metalloids

Metals occupy most of the left and centre of the table. They generally conduct heat and electricity, are malleable and ductile, and tend to lose electrons. **Nonmetals** occupy mainly the upper right and have diverse properties; many are poor electrical conductors. **Metalloids**, near the boundary, have intermediate properties and include useful semiconductors.

5.3 Periodic trends

Across a period, the nuclear charge increases while added electrons mainly enter the same shell. The increasing effective nuclear attraction generally pulls valence electrons closer. Down a group, extra shells and shielding increase the distance of valence electrons from the nucleus.

Property	Left to right across a period	Top to bottom down a group
Atomic radius	Generally decreases	Generally increases
First ionization energy	Generally increases	Generally decreases
Electronegativity	Generally increases	Generally decreases
Metallic character	Generally decreases	Generally increases

These are trends, not exact rules for every adjacent pair. Subshell structure, electron pairing and the definition of atomic radius can produce exceptions.

Ionization energy is the energy required to remove an electron from a gaseous atom or ion. The first ionization is

$$\mathrm{X(g)} \longrightarrow \mathrm{X^+(g)} + \mathrm{e^-}.$$

Successive ionizations require increasing energy; a large jump often marks the start of removing a core electron.

Electronegativity describes an atom's tendency to attract shared electrons in a bond. It is not the same as ionization energy. Fluorine has the highest electronegativity on the commonly used Pauling scale.

Cations are generally smaller than their neutral parent atoms; anions are larger. Among **isoelectronic** species (same electron count), increasing nuclear charge generally decreases radius.

5.4 Oxidation numbers

An **oxidation number** assigns bonding electrons to the more electronegative atom, as if the bond were completely ionic. It is a bookkeeping quantity and is not necessarily an actual atomic charge.

1. An element in its elemental form has oxidation number zero.

2. A monatomic ion has an oxidation number equal to its signed charge number.

3. Fluorine is -1 in its compounds. Group 1 and group 2 metals are usually $+1$ and $+2$, respectively, in their compounds.

4. Hydrogen is usually $+1$, but is -1 in metal hydrides.

5. Oxygen is usually -2; in peroxides it is -1. Oxygen bonded to fluorine can have a positive oxidation number.

6. The sum, including the number of each atom, equals the overall charge:

$$\boxed{\sum_i n_i \,\mathrm{OS}_i = q_{\text{species}}}$$

For example, sulfur in SO_4^{2-} satisfies $\mathrm{OS}_S + 4(-2) = -2$, hence $\mathrm{OS}_S = +6$. Do not apply a compound rule to a free element: oxygen in O_2 has oxidation number zero, not -2.

Element	Selected common oxidation numbers in compounds
C	$-4, +2, +4$
N	$-3, +2, +3, +4, +5$
P	$-3, +3, +5$
S	$-2, +2, +4, +6$
Cl, Br, I	$-1, +1, +3, +5, +7$
Zn	$+2$
Pb	$+2, +4$

These lists are useful common values, not exhaustive sets of possible states. Redox reactions change oxidation numbers (Appendix B).

5.5 Abundance of elements

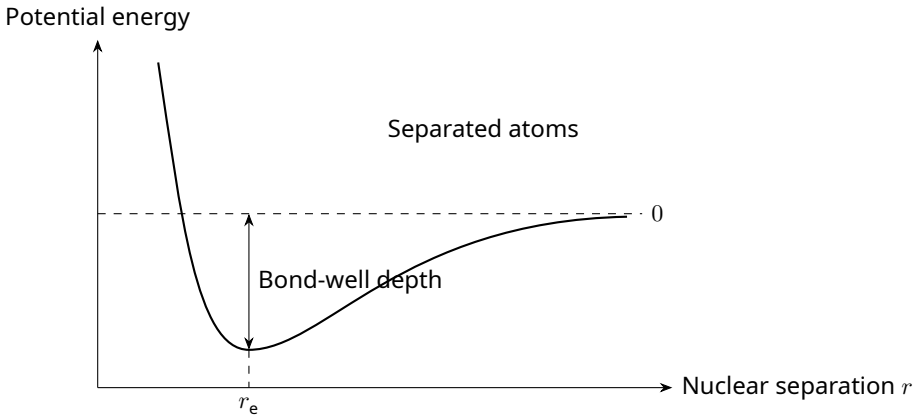
Hydrogen and helium dominate ordinary matter in the cosmos. Oxygen and silicon are major constituents of Earth's crust. Oxygen, carbon, hydrogen and nitrogen account for about 96% of human body mass. Always distinguish mass fractions from fractions of atoms when comparing elemental abundances.

6 Chemical bonds and hybridization

Course source: SW4, Chemical Bonds and Hybridization; SW5 recap.

6.1 Why atoms bond

A stable bond corresponds to a lower energy than the separated atoms under the conditions considered. Attraction between nuclei and electrons competes with electron–electron and nucleus–nucleus repulsion. The minimum of the potential-energy curve defines an equilibrium bond length.



Bond formation releases energy; separating a stable bond requires energy. The energy change of a whole reaction depends on all the bonds broken and formed.

6.2 Octet rule and its limits

Many main-group atoms form bonds or ions that give them eight valence-shell electrons, corresponding to a filled s and p shell. Hydrogen follows a **duet rule**. The octet rule is a useful pattern, not a universal law.

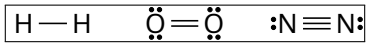
- Electron-deficient compounds such as BF_3 can have fewer than eight electrons around the central atom.
- Odd-electron species such as NO cannot give every atom an octet.
- Some period-3 and heavier central atoms have hypervalent Lewis structures, such as PCl_5 and SF_6. Period-2 atoms do not expand their octet.

Hypervalent Lewis drawings are electron-counting models; they do not require extending the simple s/p hybridization picture by assuming substantial d-orbital bonding.

6.3 Covalent, ionic and metallic bonding

6.3.1 Covalent bonding

Atoms share electron pairs. Bonding electrons are attracted by both nuclei; nonbonding valence electrons form lone pairs or, in radicals, unpaired electrons. These drawings represent single, double and



triple bonds.

6.3.2 Ionic bonding

An ionic solid is held together by electrostatic attraction between oppositely charged ions in a lattice. Electron transfer is a useful formation model:

$$\mathrm{Na} \longrightarrow \mathrm{Na^+} + \mathrm{e^-}, \qquad \mathrm{Cl} + \mathrm{e^-} \longrightarrow \mathrm{Cl^-}.$$

The formula states the smallest electrically neutral ratio of ions. For example, $MgCl_2$ contains one Mg^{2+} for every two Cl^- ions. Ionic solids often have high melting points and are brittle. Mobile ions allow conduction in the melt or, when soluble, in solution; ions in an ordinary solid lattice cannot carry current freely. Not every ionic compound is water-soluble.

6.3.3 Metallic bonding

A metal can be pictured as positive atomic cores bound by delocalized valence electrons. Mobile electrons explain electrical and thermal conductivity. Nondirectional bonding allows layers of atoms to shift, giving ductility and malleability. Metal melting points vary widely.

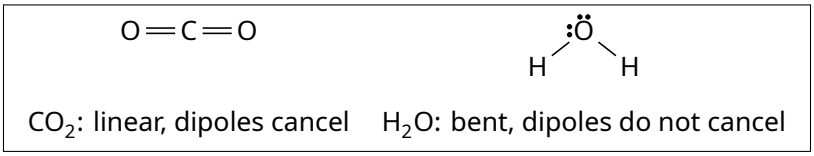
6.4 Bond polarity and molecular polarity

For bonded atoms A and B, use the electronegativity difference

$$\Delta\chi = |\chi_{\mathrm{A}} - \chi_{\mathrm{B}}|.$$

An unequal distribution of bonding electrons produces partial charges δ^+ and δ^-. The more electronegative atom carries δ^-. The lecture uses the rough guide $\Delta\chi < 1.0$ for largely nonpolar covalent, 1.0–1.6 for polar covalent and > 1.6 for predominantly ionic bonding. These cutoffs are course heuristics: bonding is continuous, and smaller nonzero differences can still produce polar bonds.

Molecular polarity depends on geometry. The molecular dipole is the vector sum of bond and other charge-distribution contributions. Polar bonds can cancel in a symmetric molecule.



CO_2: linear, dipoles cancel H_2O: bent, dipoles do not cancel

Partial charge, formal charge and oxidation number describe different things; they must not be used interchangeably.

6.5 Sigma and pi bonds

- A **sigma bond** (σ) arises from head-on orbital overlap along the internuclear axis. It can involve s, p or hybrid orbitals.
- A **pi bond** (π) arises from sideways overlap of parallel orbitals, typically unhybridized p orbitals. Its electron density lies on opposite sides of a plane containing the bond axis.

Bond order	Shared pairs	Components	Electrons shared
Single	1	1σ	2
Double	2	$1\sigma + 1\pi$	4
Triple	3	$1\sigma + 2\pi$	6

For the same pair of elements, increasing bond order generally shortens and strengthens the bond. A single sigma bond often permits rotation, although there can be torsional barriers. Rotation around a double bond disrupts pi overlap and therefore requires substantial energy.

6.6 Hybridization

Hybridization combines atomic orbitals on the **same atom** into a new set of orbitals useful for describing localized bonds. Orbital overlap between **different atoms** then forms bonds. The number of hybrid orbitals equals the number of atomic orbitals combined.

Type	Orbitals combined	Hybrid orbitals	Arrangement	Ideal angle
sp	One s and one p	Two sp; two p remain	Linear	180°
sp^2	One s and two p	Three sp^2; one p remains	Trigonal planar	120°
sp^3	One s and three p	Four sp^3; no p remains	Tetrahedral	109.5°

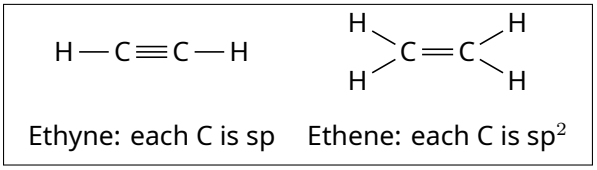
The remaining p orbitals can form pi bonds. Hybrids can contain a bonding pair, a lone pair or no electrons, depending on the species.

For simple main-group structures, the **steric number** (SN) counts electron domains around an atom: A

$$\text{SN} = (\text{number of bonded atoms}) + (\text{number of lone pairs})$$

$$\text{SN} = 2 \Rightarrow \text{sp}, \quad \text{SN} = 3 \Rightarrow \text{sp}^2, \quad \text{SN} = 4 \Rightarrow \text{sp}^3$$

single, double or triple bond to one neighbour counts as **one domain**. For extended conjugated structures, delocalization must also be considered; SN is an introductory guide rather than a universal orbital theory.



Ethyne: each C is sp Ethene: each C is sp^2

In ethyne, two unhybridized p orbitals on each carbon supply two pi bonds. In ethene, one p orbital on each carbon supplies one pi bond. In methane, four sp^3 orbitals describe four sigma bonds. Water and ammonia have four electron domains, but their lone pairs make the molecular shapes bent and pyramidal.

6.7 Coordination compounds

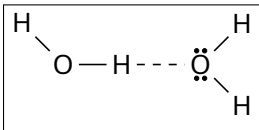
A **coordination complex** contains a central atom or ion, commonly a metal, surrounded by **ligands**. A ligand donates a lone pair to an available orbital on the centre; both electrons in the initial donor–acceptor bond come from the ligand. Once formed, the bond is covalent in character.

Common ligands include H_2O, NH_3 and Cl^-. For example, $[Cu(NH_3)_4]^{2+}$ contains neutral ammonia ligands around Cu^{2+}. The overall charge is the metal charge plus the sum of ligand charges. The **coordination number** counts donor atoms directly bonded to the centre.

6.8 Intermolecular interactions and properties

Interaction	Origin	Consequence
London dispersion	Fluctuating electron distributions induce neighbouring dipoles; present in all atoms and molecules.	Usually stronger for more polarizable particles and larger areas of contact.
Dipole–dipole	Attraction between permanent molecular dipoles.	Influences boiling points and interactions between polar molecules.
Hydrogen bonding	An H bonded to N, O or F interacts with an appropriate lone-pair acceptor, commonly N, O or F.	Often increases boiling point, surface tension and viscosity; can occur between or within molecules.
Ion–dipole	An ion interacts with a polar molecule.	Important for hydration of ions and dissolution of salts.

The term **van der Waals interactions** includes dispersion and interactions involving permanent or induced dipoles; it does not mean that every dipole is temporary. Many weak interactions acting over a large contact area can produce substantial adhesion, as in the gecko example discussed in the lecture.



Solid lines: covalent O–H bonds. Dashed line: an intermolecular hydrogen bond.

6.8.1 From bonding to macroscopic behaviour

- Molecular substances often have low melting and boiling points because phase changes overcome intermolecular attractions, not the covalent bonds inside each molecule.

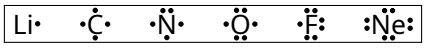
- Covalent network solids can have very high melting temperatures because strong bonds extend throughout the solid. Diamond and silica differ fundamentally from small molecular substances.
- Ionic solids, metals and molecular solids have different charge carriers; conductivity depends on whether those carriers can move.
- “Like dissolves like” summarizes a tendency for favourable interactions: polar solvents often dissolve polar or ionic solutes; nonpolar solvents often dissolve nonpolar solutes. It is not an absolute solubility rule.

7 Lewis structures and molecular geometry

Course source: SW5, Lewis Structures; MEP notes, Lewis and VSEPR procedures.

7.1 Lewis symbols

Lewis notation shows valence electrons as dots around an element symbol. A bond line stands for two shared electrons; a lone pair belongs to one atom in the localized Lewis model.



7.2 Procedure for drawing a Lewis structure

1. **Count all valence electrons.** Add one electron for each unit of negative charge; subtract one for each unit of positive charge.
2. **Choose the connectivity.** The least electronegative suitable atom is usually central. Hydrogen is always terminal; halogens are often terminal, but can be central when bonded to oxygen or other halogens.
3. **Connect atoms with single bonds.** Each bond uses two electrons.
4. **Complete terminal shells.** Give H two electrons and, where applicable, complete terminal-atom octets with lone pairs.
5. **Place remaining electrons on the central atom.** Keep the total equal to the electron count from the first step.
6. **Form multiple bonds if needed.** Convert adjacent lone pairs to bonding pairs if a central atom lacks an octet and the chemistry permits it.
7. **Check formal charges and exceptions.** Show all nonzero formal charges; enclose a polyatomic ion in brackets with its overall charge.

Electron count comes first. Do not invent extra electrons or assign an arbitrary overall charge just to complete octets. The molecular or ionic formula fixes the total number of available valence electrons.

7.3 Formal charge

Formal charge assigns half of each bond's electrons to each bonded atom, irrespective of electronegativity: V_e is the neutral atom's valence-electron count; $N_\mathrm{nonbonding}$ counts lone-pair and unpaired electrons,

$$\mathrm{FC} = V_\mathrm{e} - N_\mathrm{nonbonding} - \frac{N_\mathrm{bonding}}{2}$$

and N_bonding counts all electrons in bonds to that atom. Among reasonable structures, favour small

$$\sum_\mathrm{atoms} \mathrm{FC} = q_\mathrm{species}$$

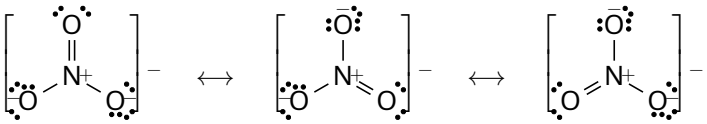
formal-charge magnitudes and minimal charge separation, with negative formal charges preferentially on more electronegative atoms. These preferences must respect the electron count and the restrictions on period-2 octets.

Quantity	Assignment of bonding electrons	Purpose
Formal charge	Shared equally between the two atoms	Compare Lewis structures.
Oxidation number	Assigned to the more electronegative atom	Track redox changes.
Partial charge	Reflects the actual electron distribution approximately	Describe bond polarity; need not be an integer.

In water, all formal charges are zero, oxygen has oxidation number -2, and the O atom carries a partial negative charge.

7.4 Resonance and delocalization

Sometimes several Lewis structures with the same atomic connectivity differ only in electron placement. They are **resonance contributors**. The real electronic structure is delocalized and is described by a resonance hybrid; the molecule does not switch back and forth between distinct structures.



Three equivalent contributors of nitrate, NO_3^-; each drawing has overall charge -1. Local formal charges and lone pairs are shown.

In each nitrate contributor, N has FC $+1$, the double-bonded O has FC 0, and each single-bonded O has FC -1. The sum is -1. The actual three N–O bonds are equivalent; the simple resonance model gives an average bond order of $(2 + 1 + 1)/3 = 4/3$. Use $\longleftrightarrow$ for resonance, and $\rightleftharpoons$ for an equilibrium between distinct chemical species.

7.5 The VSEPR model

Valence-shell electron-pair repulsion predicts arrangements that separate electron domains around a central atom. Write the notation AX_mE_n, where A is the central atom, X a bonded atom and E a lone pair on A. The steric number is $m + n$.

1. Draw a valid Lewis structure, including the overall charge.

2. Count the bonding domains and lone pairs around the central atom.

3. Assign the **electron-domain geometry** using all domains.

4. Name the **molecular geometry** using only the positions of the atoms.

Multiple bonds count as one domain, though their repulsion can differ from single bonds. A common qualitative ordering is

$$\text{lone pair–lone pair} > \text{lone pair–bonding pair} > \text{bonding pair–bonding pair}.$$

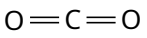
Lone pairs therefore often compress bond angles. The size of the change is species-dependent; a fixed angle reduction per lone pair is only a rough mnemonic. Charge influences the electron count and can change the geometry indirectly.

7.6 Geometry reference

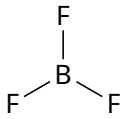
SN	Notation	Electron geometry	Molecular geometry	Bond angles	Example
2	AX_2	Linear	Linear	180°	CO_2
3	AX_3	Trigonal planar	Trigonal planar	120°	BF_3
3	AX_2E	Trigonal planar	Bent	Usually $< 120^\circ$	SO_2
4	AX_4	Tetrahedral	Tetrahedral	109.5°	CH_4
4	AX_3E	Tetrahedral	Trigonal pyramidal	About 107° in NH_3	NH_3
4	AX_2E_2	Tetrahedral	Bent	104.5° in H_2O	H_2O
5	AX_5	Trigonal bipyramidal	Trigonal bipyramidal	90°, 120°, 180°	PCl_5
5	AX_4E	Trigonal bipyramidal	Seesaw	Distorted from 90°, 120°	SF_4
5	AX_3E_2	Trigonal bipyramidal	T-shaped	About 90°, 180°	ClF_3
5	AX_2E_3	Trigonal bipyramidal	Linear	180°	XeF_2
6	AX_6	Octahedral	Octahedral	90°, 180°	SF_6
6	AX_5E	Octahedral	Square pyramidal	About 90°, 180°	BrF_5
6	AX_4E_2	Octahedral	Square planar	90°, 180°	XeF_4

For trigonal-bipyramidal electron geometry, lone pairs preferentially occupy equatorial positions, which have fewer 90° interactions. In an octahedral arrangement with two lone pairs, the pairs occupy opposite positions, leaving a square-planar arrangement of four atoms.

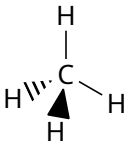
7.7 Molecular drawings



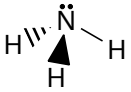
Linear: CO_2, 180°



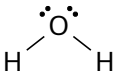
Trigonal planar: BF_3, 120°



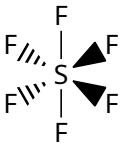
Tetrahedral: CH_4



Trigonal pyramidal: NH_3



Bent: H_2O



Octahedral: SF_6

Solid wedges point toward the viewer; hashed wedges point away. These are perspective drawings, so angles on the page are not necessarily physical bond angles. Terminal-atom lone pairs are omitted in this geometry overview.

8 Chemical reactions and stoichiometry

Course source: SW6, Chemical Reactions.

8.1 Representing a reaction

A chemical equation identifies reactants, products and their relative amounts. For example, a balanced combustion equation is The **stoichiometric coefficients** give particle and mole ratios. They do not give

$$\boxed{\mathrm{CH_4(g) + 2\,O_2(g) \longrightarrow CO_2(g) + 2\,H_2O(g)}}$$

mass ratios directly. State symbols identify solids (s), liquids (l), gases (g) and aqueous species (aq).

A one-way arrow emphasizes the net reaction in the stated direction; $\rightleftharpoons$ indicates reversibility. Temperature, catalysts or other conditions can be written above an arrow without becoming reactants.

8.2 Balancing equations

1. Write the correct chemical formulas and the reaction skeleton.
2. Count each element on both sides; also track net electrical charge.
3. Change **coefficients**, beginning with elements that occur in the fewest different species. An unchanged polyatomic ion can be treated as a unit.
4. Balance the remaining elements; hydrogen and oxygen are often convenient to leave until late in a combustion-type balance.
5. Reduce to a convenient smallest integer ratio and check every element and the total charge again.

Never change formula subscripts to balance a reaction. Changing $\mathrm{H_2O}$ to $\mathrm{H_2O_2}$ changes the substance. A coefficient changes only how many particles or moles take part.

An equation can be balanced without being spontaneous or fast. Atom balance does not establish feasibility, mechanism or rate.

8.3 Using mole ratios

For $a\mathrm{A} + b\mathrm{B} \rightarrow c\mathrm{C} + d\mathrm{D}$, complete consumption or formation in the stated stoichiometric proportions gives

$$\frac{n_{\mathrm{A},consumed}}{a} = \frac{n_{\mathrm{B},consumed}}{b} = \frac{n_{\mathrm{C},formed}}{c} = \frac{n_{\mathrm{D},formed}}{d}.$$

Convert the given mass to moles, apply the coefficient ratio, and convert back to mass if required: When

$$\boxed{m_{\mathrm{A}} \xrightarrow{\div M_{\mathrm{A}}} n_{\mathrm{A}} \xrightarrow{\times (c/a)} n_{\mathrm{C}} \xrightarrow{\times M_{\mathrm{C}}} m_{\mathrm{C}}}$$

several reactants are supplied, the smallest ratio $n_{i,0}/\nu_i$ (using positive reactant coefficients) identifies the limiting reactant for the assumed complete reaction. For equilibrium-limited reactions, stoichiometry still relates changes, but it does not imply complete consumption.

8.4 Recognizing reactions

Gas formation, a precipitate, colour changes, light emission and heat transfer can accompany reactions. No single observation proves that a chemical reaction has occurred: boiling also produces gas without changing chemical identity. Reaction classes include combination, decomposition, combustion, precipitation, acid–base reactions and redox reactions.

9 Thermodynamics and ideal gases

Course source: SW6, Chemical Reactions; formula sheet, items 3 and 7.

9.1 System and surroundings

The **system** is the matter under study; the **surroundings** are everything outside it. Together they form the universe for the energy balance.

System	Exchanges energy?	Exchanges matter?
Open	Yes	Yes
Closed	Yes	No
Isolated (ideal)	No	No

9.2 First law and sign convention

The first law expresses energy conservation. Internal energy U is a state function, whereas heat Q_{heat} and work W describe energy transfer along a process. Here $Q_{\text{heat}} > 0$ means heat enters the system,

$$\boxed{\Delta U = Q_{\text{heat}} + W_{\text{on}}}$$

and $W_{\text{on}} > 0$ means work is done **on** the system. For pressure–volume work against a constant external pressure,

$$W_{\text{on}} = -p_{\text{ext}}\Delta V.$$

Expansion gives $W_{\text{on}} < 0$; compression gives $W_{\text{on}} > 0$. If work is instead defined positive **by** the system, the equivalent form is $\Delta U = Q_{\text{heat}} - W_{\text{by}}$.

Convention used throughout these notes: work on the system is positive. The SW6 slide mixes this verbal convention with a minus sign in its displayed first law; the equations above make the convention consistent.

9.3 Ideal-gas equation and gas laws

For an ideal gas, Use absolute pressure and absolute temperature. In particular, $0\,^{\circ}\text{C} = 273.15\,\text{K}$ and

$$\boxed{pV = nRT, \qquad T[\text{K}] = \vartheta[^{\circ}\text{C}] + 273.15}$$

$1\,^{\circ}\text{C} = 274.15\,\text{K}$. Choose the gas constant to match the units:

$$R = 8.31446\,\text{J mol}^{-1}\,\text{K}^{-1} = 0.0831446\,\text{L bar mol}^{-1}\,\text{K}^{-1}.$$

Since $1\,\text{Pa m}^3 = 1\,\text{J}$, the first form works with Pa and m^3. Also, $1\,\text{atm} = 1.01325\,\text{bar}$.

For a fixed amount of ideal gas,

Conditions	Relation	Name
Constant T	$p_1V_1 = p_2V_2$	Boyle's law
Constant p	$V_1/T_1 = V_2/T_2$	Charles's law
Constant V	$p_1/T_1 = p_2/T_2$	Pressure–temperature law
General change	$p_1V_1/T_1 = p_2V_2/T_2$	Combined gas law

For an isothermal ideal-gas process, $\Delta U = 0$, so $Q_{\text{heat}} = -W_{\text{on}}$. During isothermal compression, work enters and an equal amount of heat leaves the gas.

$$\boxed{V_{\mathrm{m}} = \frac{V}{n} = \frac{RT}{p}}$$

9.4 Molar gas volume

All ideal gases have the same molar volume at the same T and p. At $273.15\,\mathrm{K}$ and $1\,\mathrm{atm}$, $V_{\mathrm{m}} \approx 22.414\,\mathrm{L\,mol^{-1}}$; at $273.15\,\mathrm{K}$ and $1\,\mathrm{bar}$, it is approximately $22.711\,\mathrm{L\,mol^{-1}}$. The formula sheet's value 22.41 corresponds to the first convention. Always state the conditions; molar volume is not constant when temperature or pressure changes. Real gases approach ideal behaviour most closely at low density and away from condensation.

9.5 Enthalpy and reaction heat

Enthalpy is defined by For a closed system at constant pressure with only pressure–volume work, $Q_p =$

$$\boxed{H = U + pV, \qquad \Delta H = \Delta U + \Delta(pV)}$$

ΔH.

- **Exothermic:** $\Delta H < 0$; the system releases heat.
- **Endothermic:** $\Delta H > 0$; the system absorbs heat.

Breaking bonds requires energy and forming bonds releases energy. Approximate reaction enthalpies can therefore be estimated from bond energies:

$$\Delta H_{\mathrm{r}} \approx \sum D(\text{bonds broken}) - \sum D(\text{bonds formed}).$$

This gas-phase average-bond-energy estimate is not a replacement for phase-specific thermochemical data.

9.6 Entropy and the second law

Entropy S reflects the distribution of energy among accessible microscopic states. "Disorder" is a useful shorthand but is not a complete definition. Gases often have higher molar entropy than corresponding liquids and solids; mixing and increasing the number of gas particles often increase entropy.

For a spontaneous process, The equality applies to the reversible limit. The system's entropy can de-

$$\boxed{\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0}$$

crease if the surroundings' increase is sufficiently large. At constant temperature of a large surroundings reservoir, heat entering the system gives $\Delta S_{\text{surroundings}} = -Q_{\text{heat}}/T$.

9.7 Gibbs energy and spontaneity

At constant temperature, For a reaction at constant T and p, the reaction Gibbs energy determines the

$$\boxed{G = H - TS, \qquad \Delta G = \Delta H - T\Delta S}$$

spontaneous direction at the **current composition**:

- $\Delta_{\mathrm{r}} G < 0$: forward reaction is thermodynamically favoured.
- $\Delta_{\mathrm{r}} G > 0$: reverse reaction is thermodynamically favoured.
- $\Delta_{\mathrm{r}} G = 0$: chemical equilibrium.

ΔH	ΔS	Condition for $\Delta G < 0$
$-$	$+$	All positive temperatures
$+$	$-$	No positive temperature
$-$	$-$	Lower temperatures: $T < \Delta H/\Delta S$
$+$	$+$	Higher temperatures: $T > \Delta H/\Delta S$

The table assumes ΔH and ΔS remain approximately constant over the temperature range. Use compatible units: convert entropy from J mol^{-1} K^{-1} to kJ mol^{-1} K^{-1} when enthalpy is in kJ mol^{-1}.

9.8 Calculations from standard data

For a balanced reaction with coefficients ν_i,

$$\Delta_\mathrm{r} H^\circ = \sum_{\text{products}} \nu_i \Delta_\mathrm{f} H_i^\circ - \sum_{\text{reactants}} \nu_i \Delta_\mathrm{f} H_i^\circ,$$

$$\Delta_\mathrm{r} S^\circ = \sum_{\text{products}} \nu_i S_i^\circ - \sum_{\text{reactants}} \nu_i S_i^\circ,$$

$$\Delta_\mathrm{r} G^\circ = \Delta_\mathrm{r} H^\circ - T \Delta_\mathrm{r} S^\circ.$$

Use each species' correct physical state and multiply by its stoichiometric coefficient. Reversing a reaction reverses the signs of its reaction quantities; multiplying the equation multiplies these quantities by the same factor.

An element in its reference state has $\Delta_\mathrm{f} H^\circ = 0$ by convention, but its standard molar entropy at room temperature is generally **not zero**. Modern standard pressure is 1 bar; many older tables and the SW6 description use 1 atm. Temperature is stated separately, commonly 298.15 K. Match the convention of the data being used.

10 Chemical equilibrium and kinetics

Course source: SW6, Chemical Reactions; formula sheet, items 5–6.

10.1 Dynamic equilibrium

In a closed reacting system that has reached chemical equilibrium, forward and reverse reaction rates are equal. Both directions continue microscopically, while macroscopic composition remains constant. Equilibrium does not require equal reactant and product concentrations, nor does every reacting mixture already satisfy equilibrium conditions.

For $a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$, the usual concentration form of the law of mass action is Square brackets denote

$$\boxed{K_c = \frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}}$$

equilibrium concentrations. Coefficients become exponents. Pure solids and pure liquids are omitted because their activities are approximately one. Solutes and gases remain in the expression.

For gases, an analogous K_p uses equilibrium partial pressures. Thermodynamic equilibrium constants are dimensionless because they use activities, such as $a_i \approx c_i/c^\circ$ for dilute ideal solutions and $a_i \approx p_i/p^\circ$ for ideal gases. Classroom K_c and K_p expressions often leave these standard-state divisors implicit. For a specified reaction and standard states, the equilibrium constant depends on temperature; it is not changed by a catalyst.

10.2 Reaction quotient and direction

The **reaction quotient** Q_r has the same form as K, but uses the current composition, whether or not equilibrium has been reached.

Comparison	Net tendency
$Q_\mathrm{r} < K$	Forward, toward products
$Q_\mathrm{r} > K$	Reverse, toward reactants
$Q_\mathrm{r} = K$	At equilibrium

The thermodynamic connection is These equations require dimensionless activity-based Q_r and K. A

$$\boxed{\Delta_\mathrm{r}G = \Delta_\mathrm{r}G^\circ + RT\ln Q_\mathrm{r}, \qquad \Delta_\mathrm{r}G^\circ = -RT\ln K}$$

large K indicates product-favoured equilibrium, not a fast reaction. A negative $\Delta_\mathrm{r}G^\circ$ describes standard-state driving force; the actual driving force also depends on composition.

10.3 Le Chatelier's principle

When an equilibrium system is disturbed, its composition tends to change in a direction that partly counteracts the disturbance.

Change	Equilibrium response
Add a reactant	Tends to consume it, forming more products.
Remove a product	Tends to replace it, forming more products.
Compress an ideal gas mixture at fixed T	Favours the side with fewer gas moles; no such shift if gas stoichiometric totals are equal.
Increase temperature	Favours the endothermic direction; changes K.
Decrease temperature	Favours the exothermic direction; changes K.
Add a catalyst	Reaches equilibrium faster; does not change K or the equilibrium composition.
Add more of a pure solid already present	No change in its activity, so no equilibrium shift solely from its increased amount.

Only gaseous species count in the compression rule. Adding an inert gas at constant volume leaves the reacting gases' partial pressures unchanged in the ideal-gas approximation; increasing total pressure this way does not shift the equilibrium. Changing temperature differs from changing concentration: it changes the value of K itself.

10.4 Reaction rate

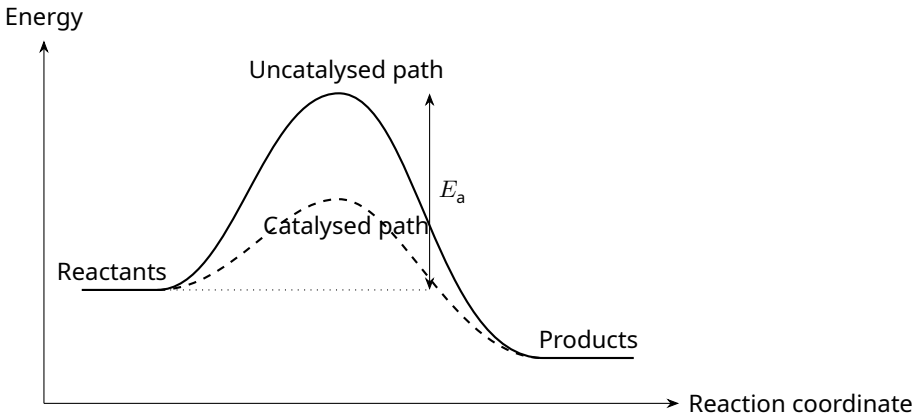
Kinetics studies how fast a process proceeds. At constant volume, a normalized average rate for the general reaction is The common unit is mol L^{-1} s^{-1}. For the instantaneous rate, replace finite changes by

$$\boxed{v = -\frac{1}{a}\frac{\Delta[\mathrm{A}]}{\Delta t} = -\frac{1}{b}\frac{\Delta[\mathrm{B}]}{\Delta t} = \frac{1}{c}\frac{\Delta[\mathrm{C}]}{\Delta t} = \frac{1}{d}\frac{\Delta[\mathrm{D}]}{\Delta t}}$$

derivatives. The negative sign for reactants makes their disappearance correspond to a positive forward rate. Stoichiometric coefficients relate species rates; they do not generally give the exponents in an experimentally determined rate law.

10.5 Activation energy and catalysts

A thermodynamically favoured reaction can be slow because it must pass through a high-energy transition region. The **activation energy** measures the kinetic barrier in an energy-profile description.



A **catalyst** provides another mechanism with a lower effective barrier. It participates in steps but is regenerated overall. It accelerates approach to equilibrium in both directions and does not alter the reaction's overall ΔG, ΔH or equilibrium constant.

Rates can often be increased by raising temperature, increasing reactant concentration or gas partial pressure, improving mixing, or increasing the exposed surface area of a solid reactant. The effect depends on the mechanism.

Three separate questions: stoichiometry describes the allowed amount ratios; thermodynamics describes the favoured direction and equilibrium; kinetics describes the rate and pathway.

A Acid–base reference from the supplied sheets

Course source: formula sheet, items 8–17; MEP notes, acid-constant table.

This appendix explains the acid–base formulas already present in the supplied reference sheets. No separate acid–base lecture slides were present in the source folder at the time these notes were prepared.

A.1 Acids, bases and conjugate pairs

In the Brønsted–Lowry description, an acid donates a proton and a base accepts one. A conjugate acid–base pair differs by one proton:

$$\mathrm{HA} + \mathrm{H_2O} \rightleftharpoons \mathrm{H_3O^+} + \mathrm{A^-}, \qquad \mathrm{B} + \mathrm{H_2O} \rightleftharpoons \mathrm{BH^+} + \mathrm{OH^-}.$$

$\mathrm{HA/A^-}$ and $\mathrm{BH^+/B}$ are conjugate pairs. Water can act as an acid or a base. In aqueous solution, $\mathrm{H^+}$ is often used as shorthand for a solvated proton, represented here by $\mathrm{H_3O^+}$.

A.2 Water, pH and pOH

$$2\,\mathrm{H_2O} \rightleftharpoons \mathrm{H_3O^+} + \mathrm{OH^-}.$$

The rigorous definitions use dimensionless activities: For dilute ideal solutions, use $a_i \approx [i]/c^\circ$, with

$$\boxed{K_\mathrm{w} = a(\mathrm{H_3O^+})a(\mathrm{OH^-}), \qquad \mathrm{pH} = -\log_{10} a(\mathrm{H_3O^+}), \qquad \mathrm{pOH} = -\log_{10} a(\mathrm{OH^-})}$$

$c^\circ = 1\,\mathrm{mol\,L^{-1}}$. The familiar concentration approximation at $25\,^\circ\mathrm{C}$ is

$$[\mathrm{H_3O^+}][\mathrm{OH^-}] \approx 10^{-14}\,\mathrm{mol^2\,L^{-2}}, \qquad \mathrm{pH} + \mathrm{pOH} = \mathrm{p}K_\mathrm{w} \approx 14.$$

Neutrality means equal hydronium and hydroxide activities. Therefore neutral pH is $\mathrm{p}K_\mathrm{w}/2$, which is approximately 7 at $25\,^\circ\mathrm{C}$ but changes with temperature.

A.3 Acid and base constants

In the usual dilute-solution notation (standard-state factors implicit), The course sheet uses K_S for K_a

$$\boxed{K_\mathrm{a} \approx \frac{[\mathrm{H_3O^+}][\mathrm{A^-}]}{[\mathrm{HA}]}, \qquad K_\mathrm{b} \approx \frac{[\mathrm{BH^+}][\mathrm{OH^-}]}{[\mathrm{B}]}}$$

and K_B for K_b. With a consistent standard-state convention,

$$\mathrm{p}K_\mathrm{a} = -\log_{10} K_\mathrm{a}, \qquad \mathrm{p}K_\mathrm{b} = -\log_{10} K_\mathrm{b}.$$

A larger acid constant means a stronger acid and a smaller $\mathrm{p}K_\mathrm{a}$. A stronger acid has a weaker conjugate base. For a conjugate pair at the same temperature, Acid strength is an equilibrium property; it is

$$\boxed{K_\mathrm{a}K_\mathrm{b} = K_\mathrm{w}, \qquad \mathrm{p}K_\mathrm{a} + \mathrm{p}K_\mathrm{b} = \mathrm{p}K_\mathrm{w}}$$

different from concentration. Strong acids dissociate essentially completely in the usual dilute aqueous regime. Weak acids establish an appreciable undissociated fraction.

A.4 Useful pH approximations

Let c_0 be the analytical concentration, and define the dimensionless value $\tilde{c}_0 = c_0/c^\circ$ for use inside logarithms. The following expressions assume ideal dilute solutions and negligible water autoionization relative to the acid or base contribution.

Case	Leading approximation	pH relation
Strong monoprotic acid	$[\mathrm{H_3O^+}] \approx c_0$	$\mathrm{pH} \approx -\log_{10} \tilde{c}_0$
Weak monoprotic acid	$[\mathrm{H_3O^+}]/c^\circ \approx \sqrt{K_\mathrm{a}\tilde{c}_0}$	$\mathrm{pH} \approx \frac{1}{2}(\mathrm{p}K_\mathrm{a} - \log_{10} \tilde{c}_0)$
Strong base giving one $\mathrm{OH^-}$ per formula unit	$[\mathrm{OH^-}] \approx c_0$	$\mathrm{pH} \approx \mathrm{p}K_\mathrm{w} + \log_{10} \tilde{c}_0$
Weak monobasic base	$[\mathrm{OH^-}]/c^\circ \approx \sqrt{K_\mathrm{b}\tilde{c}_0}$	$\mathrm{pH} \approx \mathrm{p}K_\mathrm{w} - \frac{1}{2}(\mathrm{p}K_\mathrm{b} - \log_{10} \tilde{c}_0)$

For a completely dissolved strong base supplying z hydroxide ions per formula unit, replace c_0 by zc_0 in the hydroxide calculation. Polyprotic acids require attention to successive dissociation constants; not all protons can automatically be counted as completely dissociated.

A.4.1 Checking the weak-acid approximation

For a monoprotic acid without added conjugate base, set $x = [\mathrm{H_3O^+}]/c^\circ$. Neglecting water autoionization,

$$K_\mathrm{a} = \frac{x^2}{\tilde{c}_0 - x}, \qquad x = \frac{-K_\mathrm{a} + \sqrt{K_\mathrm{a}^2 + 4K_\mathrm{a}\tilde{c}_0}}{2}.$$

The square-root shortcut assumes $x \ll \tilde{c}_0$; check the fraction $x/\tilde{c}_0$ afterwards (about 5% or less is a common working criterion). Extremely dilute solutions require including water autoionization. The analogous calculation for a weak base uses K_b and hydroxide concentration.

A.5 Buffers

A buffer contains significant amounts of a weak acid and its conjugate base. Added acid is consumed mainly by the base component; added base is consumed mainly by the acid component. The Henderson–Hasselbalch relation is For equal acid and base activities, $\mathrm{pH} = \mathrm{p}K_\mathrm{a}$. Concentration is a suitable substi-

$$\boxed{\mathrm{pH} = \mathrm{p}K_\mathrm{a} + \log_{10}\left(\frac{a(\mathrm{A^-})}{a(\mathrm{HA})}\right) \approx \mathrm{p}K_\mathrm{a} + \log_{10}\left(\frac{[\mathrm{A^-}]}{[\mathrm{HA}]}\right)}$$

tute when activity effects are small. After adding strong acid or base, first update the amounts by the neutralization stoichiometry, then apply the buffer relation if both components remain. Dilution approximately preserves their ratio but reduces buffer capacity.

A.6 Selected acid data

The following representative entries are reproduced numerically from the MEP reference table; use a consistent set of constants for a calculation.

Acid	Conjugate base	K_a	$\mathrm{p}K_\mathrm{a}$
HSO_4^-	SO_4^{2-}	1.2×10^{-2}	1.92
H_3PO_4	$H_2PO_4^-$	7.52×10^{-3}	2.12
HF	F^-	7.2×10^{-4}	3.14
HCOOH	$HCOO^-$	1.77×10^{-4}	3.75
CH_3COOH	CH_3COO^-	1.76×10^{-5}	4.75
H_2CO_3 (table convention)	HCO_3^-	4.3×10^{-7}	6.37
$H_2PO_4^-$	HPO_4^{2-}	6.23×10^{-8}	7.21
HCN	CN^-	6.17×10^{-10}	9.21
NH_4^+	NH_3	5.6×10^{-10}	9.25
HCO_3^-	CO_3^{2-}	4.8×10^{-11}	10.32

The common tabulated first “carbonic acid” value uses the dissolved CO_2/H_2CO_3 pool convention; it should not be confused with dissociation of isolated molecular H_2CO_3 alone. The original MEP table contains a much larger selection of acids and alternative tabulations.

B Redox and electrochemical reference

Course source: MEP notes, oxidation numbers, redox balancing and potential tables; formula sheet, item 18.

This appendix summarizes the redox procedures and electrode-potential data provided as reference material. It does not assume the availability of later redox lecture slides.

B.1 Oxidation and reduction

Oxidation is electron loss and increases oxidation number. **Reduction** is electron gain and decreases oxidation number. They occur together because electrons transferred must be conserved.

- The **reducing agent** donates electrons and is oxidized.
- The **oxidizing agent** accepts electrons and is reduced.

$$\boxed{\mathrm{Zn} \longrightarrow \mathrm{Zn^{2+}} + 2\,\mathrm{e^-} \qquad \mathrm{Cu^{2+}} + 2\,\mathrm{e^-} \longrightarrow \mathrm{Cu}}$$

In this illustration, zinc is the reducing agent and Cu^{2+} the oxidizing agent. The overall reaction is Zn + $Cu^{2+} \longrightarrow Zn^{2+}$ + Cu. Assign oxidation numbers using Section 5.4; distinguish these from formal charges in Lewis structures.

B.2 Balancing redox equations by half-reactions

1. Write the skeleton equation and identify which oxidation numbers change.
2. Split the process into oxidation and reduction half-reactions.
3. Balance all atoms other than H and O in each half-reaction.
4. In **acidic solution**, balance O with H_2O, then H with H^+.
5. Balance electrical charge with electrons. Oxidation produces electrons; reduction consumes electrons.
6. Multiply the half-reactions so they exchange the same number of electrons. Add them and cancel electrons and identical species appearing on both sides.
7. For **basic solution**, add enough OH^- to both sides to consume every H^+; combine $H^+ + OH^- \longrightarrow H_2O$ and cancel excess water.
8. Check that every atom and the total charge are balanced and that the equation is consistent with the stated medium.

The MEP sheet uses H_3O^+. The H^+ convention above is equivalent: replace each H^+ by H_3O^+ and add one H_2O to the opposite side, then cancel water if possible. In a final basic-medium equation, no free H^+ should remain.

B.3 Standard reduction potentials

A standard reduction potential E° describes a half-reaction written as a **reduction**, relative to the standard hydrogen electrode. The MEP tables list reduced and oxidized forms in separate columns; reconstruct the balanced reduction half-reaction before using a number.

Reduction half-reaction	E° [V] (MEP values)
$\mathrm{Na^+ + e^- \longrightarrow Na}$	-2.71
$\mathrm{Mg^{2+} + 2\,e^- \longrightarrow Mg}$	-2.36
$\mathrm{Al^{3+} + 3\,e^- \longrightarrow Al}$	-1.66
$\mathrm{Zn^{2+} + 2\,e^- \longrightarrow Zn}$	-0.76
$\mathrm{Fe^{2+} + 2\,e^- \longrightarrow Fe}$	-0.41
$\mathrm{2\,H^+ + 2\,e^- \longrightarrow H_2}$	0.00
$\mathrm{Cu^{2+} + 2\,e^- \longrightarrow Cu}$	$+0.35$
$\mathrm{Fe^{3+} + e^- \longrightarrow Fe^{2+}}$	$+0.77$
$\mathrm{Ag^+ + e^- \longrightarrow Ag}$	$+0.80$
$\mathrm{Cl_2 + 2\,e^- \longrightarrow 2\,Cl^-}$	$+1.36$
$\mathrm{MnO_4^- + 8\,H^+ + 5\,e^- \longrightarrow Mn^{2+} + 4\,H_2O}$	$+1.51$

These are the rounded values in the supplied table at $298.15\,\mathrm{K}$; other data tables can differ slightly. A more positive reduction potential means a greater tendency for the oxidized form to accept electrons under the specified conditions. Actual potentials depend on activities and, where $\mathrm{H^+}$ or $\mathrm{OH^-}$ occurs, on pH.

B.4 Cells and the Nernst equation

Oxidation occurs at the **anode**; reduction occurs at the **cathode**. Using reduction potentials for both half-cells, z is the number of electrons transferred in the balanced overall reaction, and $F \approx 96485\,\mathrm{C\,mol^{-1}}$ is

$$\boxed{E^\circ_{\mathrm{cell}} = E^\circ_{\mathrm{cathode}} - E^\circ_{\mathrm{anode}}, \qquad \Delta_{\mathrm{r}}G^\circ = -zFE^\circ_{\mathrm{cell}}}$$

the Faraday constant. A positive cell potential corresponds to a negative reaction Gibbs energy for the direction written. Do **not** multiply electrode potentials by stoichiometric coefficients when equalizing electron counts: potentials are intensive quantities.

For a half-reaction written as a reduction, or a balanced full-cell reaction, Q_{r} is the dimensionless activity

$$\boxed{E = E^\circ - \frac{RT}{zF}\ln Q_{\mathrm{r}}}$$

quotient for that reaction; electrons are omitted, and pure solids have unit activity. At $25\,^\circ\mathrm{C}$,

$$E = E^\circ - \frac{0.025693\,\mathrm{V}}{z}\ln Q_{\mathrm{r}} = E^\circ - \frac{0.05916\,\mathrm{V}}{z}\log_{10} Q_{\mathrm{r}}.$$

For the simple half-reaction $\mathrm{Ox} + z\mathrm{e^-} \rightleftharpoons \mathrm{Red}$, this becomes the form used in the course sheet: The ratio

$$\boxed{E = E^\circ + \frac{RT}{zF}\ln\left(\frac{a_{\mathrm{Ox}}}{a_{\mathrm{Red}}}\right)}$$

must include all reacting species and their stoichiometric powers when a more complicated half-reaction is used. The simplified concentration ratio alone is insufficient, for example, for an acidic permanganate reaction.

C Formula overview

Topic	Relation and reminder
Density	$\rho = m/V$. Use the density at the stated conditions.
Amount of substance	$n = m/M = N/N_\mathrm{A}$. Specify the counted entities.
Molarity	$c = n/V_\mathrm{solution} = m/(MV_\mathrm{solution})$.
Mass concentration	$\beta = m/V = cM$.
Dilution	$c_1V_1 = c_2V_2$ when solute amount is conserved.
Nuclide counts	$N_\mathrm{n} = A - Z$, $N_\mathrm{e} = Z - q$.
Radioactive decay	$N = N_0e^{-\lambda t}$, $t_{1/2} = \ln 2/\lambda$, $\mathcal{A} = \lambda N$.
Orbital capacity	2 electrons per orbital; $2(2\ell + 1)$ per subshell; $2n^2$ per shell.
Formal charge	$\mathrm{FC} = V_\mathrm{e} - N_\mathrm{nonbonding} - N_\mathrm{bonding}/2$.
Electron domains	$\mathrm{SN} =$ bonded neighbours $+$ central lone pairs.
Ideal gas	$pV = nRT$ and $V_\mathrm{m} = RT/p$; T in kelvin.
First law	$\Delta U = Q_\mathrm{heat} + W_\mathrm{on}$; $W_\mathrm{on} = -p_\mathrm{ext}\Delta V$ at constant external pressure.
Enthalpy	$H = U + pV$; $\Delta H = Q_p$ under the stated constant-pressure conditions.
Gibbs energy	$\Delta G = \Delta H - T\Delta S$ at constant T.
Reaction from tables	$\Delta_\mathrm{r}X^\circ = \sum_\mathrm{prod} \nu_i X_i^\circ - \sum_\mathrm{react} \nu_i X_i^\circ$, using formation data for H and G, and absolute standard values for S.
Equilibrium	$K_c = [\mathrm{C}]^c[\mathrm{D}]^d/([\mathrm{A}]^a[\mathrm{B}]^b)$; evaluate at equilibrium and omit pure solids and liquids.
Driving force	$\Delta_\mathrm{r}G = \Delta_\mathrm{r}G^\circ + RT\ln Q_\mathrm{r}$; $\Delta_\mathrm{r}G^\circ = -RT\ln K$.
Reaction rate	$v = -(1/a)\,\Delta[\mathrm{A}]/\Delta t = (1/c)\,\Delta[\mathrm{C}]/\Delta t$ at constant volume.
Acid–base pair	$K_\mathrm{a}K_\mathrm{b} = K_\mathrm{w}$; $\mathrm{p}K_\mathrm{a} + \mathrm{p}K_\mathrm{b} = \mathrm{p}K_\mathrm{w}$.
pH and pOH	$\mathrm{pH} = -\log_{10} a(\mathrm{H_3O^+})$; $\mathrm{pH} + \mathrm{pOH} = \mathrm{p}K_\mathrm{w}$, approximately 14 at $25\,^\circ\mathrm{C}$.
Buffer	$\mathrm{pH} \approx \mathrm{p}K_\mathrm{a} + \log_{10}([\mathrm{A^-}]/[\mathrm{HA}])$.
Cell potential	$E^\circ_\mathrm{cell} = E^\circ_\mathrm{cathode} - E^\circ_\mathrm{anode}$ using reduction potentials.
Nernst	$E = E^\circ - (RT/zF)\ln Q_\mathrm{r}$.

D Source map and conventions

The local course PDFs below supply the content and topic order. Page numbers refer to **PDF page positions**, which can differ from the numbers printed on the slides. All paths are relative to `TA.BA_CHEM_MM.H2602` beside this document.

1. **SW1: Introduction to Substances.**

   `SW01IntroductiontoSubstances(Matter)/LectureMaterialandExercises/CHEM_SW1_IntroductiontoSubstances.pdf`

   Theory on PDF pp. 11–20, 23–47; supports Sections 1 and 2. Exercise activities are omitted even though the lecture's parent folder includes "Exercises" in its name.

2. **SW2: Structure of atoms.**

   `SW2_AtomicStructureandRadioactivity/SW2_Structureofatoms.pdf`

   Theory on PDF pp. 5–16 and 18–34; supports Section 3. The elementary-particle excursus is not developed further.

3. **SW3: Atomic models, periodic table and electronic configuration.**

   `SW3_AtomicModels(SelfStudy),Periodictableandelectronicconfiguration/SW3_Periodictable_electronicconfLecture.pdf`

   Self-study models on PDF pp. 2–8; periodic-table theory on pp. 13–24, 27–30 and 33–35; configurations on pp. 44–60 and conceptual reminders on p. 67. Supports Sections 4 and 5. Embedded exercise and answer pages are excluded.

4. **SW4: Chemical Bonds and Hybridization.**

   `SW4_ChemicalBondsandHybridization/CHEM_SW6_ChemicalBonds_new.pdf`

   Theory on PDF pp. 8–21, 26–28, 30–36 and 39–41; supports Section 6. The file is named "SW6", but belongs to the SW4 folder.

5. **SW5: Lewis Structures.**

   `SW5_LewisStructure/CHEM_SW5_Lewis-ILIAS.pdf`

   Theory and recap on PDF pp. 7–12, 16–17, 21 and 23–26; supports Sections 6 and 7.

6. **SW6: Chemical Reactions.**

   `SW6_ChemicalReactions/CHEM_SW6_ChemicalReactions.pdf`

   Theory on PDF pp. 5–8, 10–14 and 16–23; supports Sections 8–10. Exercise prompts and worked exercise tasks are omitted, including those sharing theory pages.

7. **Course formula sheet.**

   `Formulas/CHEM_Formulas.pdf`

   Both pages: particle counts, molar quantities, rates, equilibrium, Gibbs energy, acid–base relations and Nernst equation.

8. **MEP reference notes.**

   `MEPMaterial/MEPNotesCHEM.pdf`

   PDF pp. 2–3: Lewis and VSEPR procedures; p. 4: periodic-table reference; pp. 5–7: oxidation numbers and redox balancing; p. 8: acid constants; pp. 9–10: electrode potentials. Selected data are included in Appendices A and B; the original PDF retains the full reference tables.

Clarifications applied to the course material

Definitions and formulas are written consistently rather than retaining obvious slide transcription errors. In particular:

- The mole counts a fixed number of specified entities; it is not the number of particles in one gram. The definition follows the BIPM SI definition of the mole.

- Standard pressure is distinguished from atmospheric pressure, following the IUPAC definition of standard pressure. Gas molar volumes always have explicit temperature and pressure conditions.
- The first-law work sign is consistent; Celsius-to-kelvin conversion includes the full temperature offset, and the gas constant includes its volume unit.
- Formal charge and oxidation number are distinct. Electron counts include ionic charge before determining geometry; lone-pair angle changes are qualitative.
- Equilibrium is a condition reached by a system, not a property of every instantaneous reacting mixture. Spontaneity is distinguished from reaction rate.
- The pOH sign and base-constant subscript in the formula sheet are corrected. Logarithms use dimensionless activities or explicitly normalized concentrations, following the IUPAC pH definition.
- Nernst expressions use a balanced reduction reaction and its full activity quotient, consistent with the IUPAC Green Book electrochemical convention.